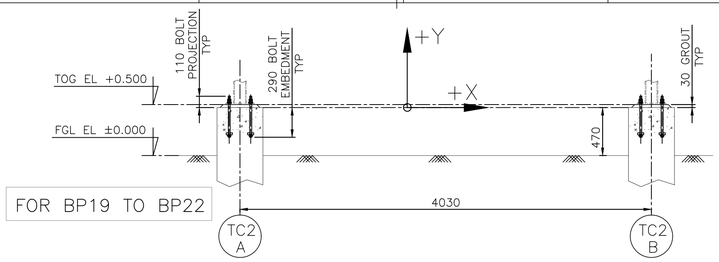


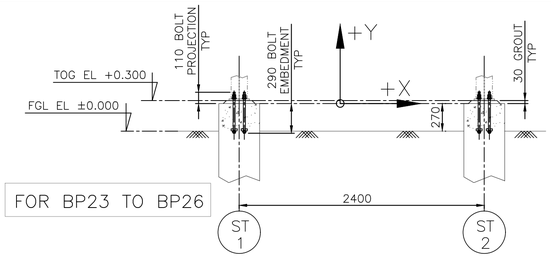
FOR BP15 TO BP18

SECTION D
SCALE 1:30



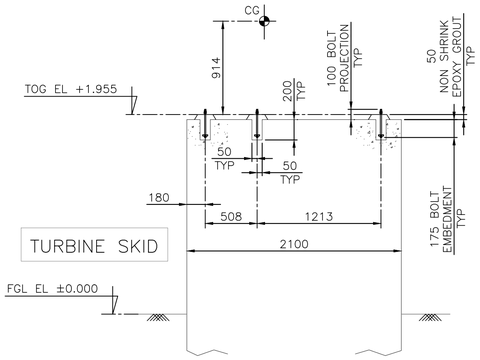
FOR BP19 TO BP22

SECTION E
SCALE 1:30



FOR BP23 TO BP26

SECTION G
SCALE 1:30



SECTION F
SCALE 1:30

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETER.
2. ALL LEVELS ARE IN METER.
3. FOR ALL OTHER NOTES SEE SHEET 1 OF 2.
4. SIZE OF FOUNDATION AND PEDESTAL SHALL BE DECIDED BY CIVIL DESIGN ENGINEER AS PER GIVEN LOADING.
5. WIND & SEISMIC LOADS ARE AS PER SBC 301.

TURBINE SKID LOAD DATA

LOAD AT CG

OPERATING LOAD – 17.8 KN

WIND LOAD :
HOZ X – 3.5 KN
HOZ Z – 6.5 KN

LIMIT TRANSMISSION OF AMBIENT VIBRATION TO LESS THAN 2.0 MILS [50.8uM] AMPLITUDE OVER THE RANGE FROM 30 Hz TO 10,000 Hz.

FOUNDATION LOAD : COLD BOX

SL No	Description	BP1			BP2			BP3			BP4			BP5/ BP6 /BP7 /BP8			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	-27	-192	-5	-32	-189	4	31	-177	21	25	-155	-21	0	-9	0	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS. B. PIPE LOAD, INSULATION AND EQUIPMENT LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY. C. INSULATION LOAD SHALL NOT BE ADDED FOR ERECTION LOAD COMBINATIONS. D. FOR REVERSE DIRECTION OF LOAD CASE 8, 9, 14 AND 15, MULTIPLY ALL VALUES WITH -1.
2	INSULATION LOAD	-14	-90	-2	-17	-83	2	16	-94	13	13	-89	-13	0	-29	0	
3	EQUIPMENT EMPTY LOAD	-5	-51	-1	-4	-18	1	4	-24	7	5	-47	-7	0	-65	0	
4	EQUIPMENT OPERATING LOAD	-5	-53	-1	-5	-19	1	5	-27	9	5	-49	-9	0	-92	0	
5	PIPE EMPTY LOAD	-8	-52	-2	-3	9	1	3	-15	7	8	-88	-7	0	-14	0	
6	PIPE OPERATING LOAD	-10	-67	-2	-4	9	1	4	-12	7	9	-102	-7	0	-16	0	
7	LIVE LOAD	-7	-109	-10	-8	-107	10	7	-21	3	7	-13	-3	0	0	0	
8	PIPE LOAD +X	7	25	-1	6	25	1	7	-24	-1	6	-25	2	0	-1	0	
9	PIPE LOAD +Z	-3	1	5	1	-1	6	10	-79	8	-9	78	23	0	-1	0	
10	WIND LOAD +X	144	892	6	108	875	-8	89	-874	62	90	-891	-60	0	-9	0	
11	WIND LOAD -X	-125	-759	-3	-87	-723	7	-66	721	-53	-78	758	50	0	-5	0	
12	WIND LOAD +Z	-3	367	40	-30	-360	43	147	-1203	76	-123	1191	216	0	-8	0	
13	WIND LOAD -Z	8	-354	-39	32	359	-41	-145	1148	-70	117	-1147	-201	0	-13	0	
14	SEISMIC LOAD +X	16	127	1	12	122	-2	8	-122	10	9	-127	-9	0	-2	0	
15	SEISMIC LOAD +Z	4	75	4	-9	-74	4	19	-168	10	-15	167	27	0	-1	0	

NOTE: PIPE LOAD +X & PIPE LOAD+Z ARE THE HORIZONTAL LOADS GENERATED DUE TO PIPE STRESS. THESE TWO LOADS SHALL BE CONSIDERED ACTING SIMULTANEOUSLY IN OPERATING CONDITION.

FOUNDATION LOAD : PUMP CASING

SL No	Description	BP9			BP10			BP11			BP12			BP13			BP14			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	-0.3	-3.4	0.0	0.4	-6.3	0.1	0.0	-0.5	0.1	-0.2	-4.5	0.0	0.2	-5.6	-0.2	0.0	-1.8	0.0	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS. B. PIPE LOAD AND EQUIPMENT LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY. C. FOR REVERSE DIRECTION OF LOAD CASE 6 TO 11, MULTIPLY ALL VALUES WITH -1.
2	EQUIPMENT EMPTY LOAD	0.0	0.2	0.0	0.0	-1.1	0.1	0.0	-1.2	0.1	0.0	-0.2	0.0	0.0	-2.3	-0.1	0.0	-2.3	-0.1	
3	PIPE EMPTY LOAD	-0.4	-1.4	0.0	0.4	-3.4	0.0	0.0	0.8	0.0	-0.2	-4.6	0.0	0.2	-4.2	-0.2	0.0	0.0	0.0	
4	PIPE OPERATING LOAD	-0.6	-2.6	0.0	0.6	-5.8	0.2	0.0	1.4	0.0	-0.4	-7.8	0.0	0.4	-7.2	-0.4	0.0	0.0	0.0	
5	LIVE LOAD	-0.4	-3.8	0.1	0.4	-5.7	0.2	0.0	-0.3	0.2	-0.2	-5.4	-0.1	0.2	-6.9	-0.4	0.0	-1.8	0.0	
6	PIPE LOAD +X	1.1	1.3	0.2	1.5	1.3	0.0	1.2	-2.6	-0.2	0.8	0.4	0.2	1.0	0.9	0.0	0.9	-1.3	-0.3	
7	PIPE LOAD +Z	0.0	-3.0	-1.0	0.1	-1.5	-0.7	0.1	-1.1	-0.3	-0.1	3.0	-1.1	-0.1	1.5	-0.6	-0.1	1.1	-0.3	
8	WIND LOAD +X	1.2	0.9	0.0	1.6	1.6	0.0	1.3	-2.5	0.0	0.8	1.0	0.0	1.0	0.7	0.0	0.9	-1.7	-0.2	
9	WIND LOAD +Z	0.0	8.5	2.5	-0.3	6.3	2.2	-0.2	4.7	1.6	0.0	-8.7	2.8	0.1	-6.2	2.1	0.0	-4.6	1.6	
10	SEISMIC LOAD +X	0.2	0.3	0.0	0.3	0.3	0.0	0.2	-0.6	0.0	0.2	0.2	0.0	0.2	0.2	0.0	0.2	-0.4	0.0	
11	SEISMIC LOAD +Z	0.0	1.0	0.2	0.0	0.8	0.2	0.0	0.6	0.2	0.0	-1.0	0.3	0.0	-0.8	0.2	0.0	-0.6	0.2	

NOTE: PIPE LOAD +X & PIPE LOAD+Z ARE THE HORIZONTAL LOADS GENERATED DUE TO PIPE STRESS. THESE TWO LOADS SHALL BE CONSIDERED ACTING SIMULTANEOUSLY IN OPERATING CONDITION.

FOUNDATION LOAD : STAIR TOWER

SL No	Description	BP23			BP24			BP25			BP26			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	-1.3	-8.2	-0.3	-0.2	-5.9	0.3	0.2	-5.8	0.3	1.3	-7.3	-0.3	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS. B. FOR REVERSE DIRECTION OF LOAD CASE 3 TO 6, MULTIPLY ALL VALUES WITH -1.
2	LIVE LOAD	-3.7	-15.1	-0.7	-0.1	-11.6	0.7	0.1	-11.4	0.8	3.7	-12.2	-0.8	
3	WIND LOAD +X	12.1	27.2	1.3	8.4	19.2	0.0	8.4	-19.2	-0.1	12.3	-27.2	-1.2	
4	WIND LOAD +Z	0.0	5.9	7.8	0.3	-6.3	5.5	0.4	-9.4	4.2	-0.7	9.2	10.8	
5	SEISMIC LOAD +X	0.3	1.0	0.0	0.3	0.7	0.0	0.3	-0.7	0.0	0.3	-1.0	0.0	
6	SEISMIC LOAD +Z	0.0	0.3	0.2	0.0	-0.3	0.2	0.0	-0.3	0.1	0.0	0.3	0.3	

FOUNDATION LOAD : TURBINE CASING

SL No	Description	BP15			BP16			BP17			BP18			BP19			BP20			BP21			BP22			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	-1.8	-12.4	-5.7	-2.6	-24.2	5.4	2.8	-17.9	3.7	1.5	-8.9	-3.5	-4.7	-14.8	0.4	-1.7	-15.1	-0.1	0.6	-8.7	-0.1	5.8	-22.1	-0.2	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS. B. PIPE LOAD LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY. C. FOR REVERSE DIRECTION OF LOAD CASE 5 TO 10, MULTIPLY ALL VALUES WITH -1.
2	PIPE EMPTY LOAD	-0.8	-2.7	-1.2	-0.7	-4.9	1.2	0.7	-4.9	1.4	0.8	-2.7	-1.4	-3.9	-12.3	0.3	-2.0	-12.2	0.0	1.2	1.4	0.0	4.7	-18.3	-0.3	
3	PIPE OPERATING LOAD	-0.8	-3.1	-1.6	-1.6	-6.9	1.4	1.6	-10.2	2.4	0.8	-3.9	-2.4	-6.1	-15.3	0.3	-4.3	-15.2	0.0	3.4	-2.2	0.0	6.9	-21.8	-0.3	
4	LIVE LOAD	-2.2	-13.3	-6.8	-2.5	-30.8	6.7	3.0	-18.2	4.9	1.6	-9.3	-4.4	-4.0	-13.9	0.4	-1.4	-20.2	-0.5	0.1	-21.5	-0.5	5.3	-21.2	0.2	
5	PIPE LOAD +X	3.4	7.0	0.2	3.2	7.2	0.4	3.7	-8.5	0.5	3.6	-5.7	1.1	-2.4	-9.7	-2.0	-1.2	-0.9	0.1	-1.6	4.4	0.0	-2.3	6.1	-0.2	
6	PIPE LOAD +Z	0.2	-0.5	-0.6	-0.2	0.5	-0.8	-0.2	0.2	0.1	0.2	-0.2	0.1	0.9	-30.7	-9.4	-0.7	30.7	0.0	-1.9	1.7	0.0	1.7	-1.7	0.7	
7	WIND LOAD +X	5.1	14.1	3.7	11.7	21.4	1.4	11.8	-22.8	0.5	5.2	-12.3	-2.7	8.0	9.9	0.8	7.7	11.1	0.0	8.3	-10.6	0.0	6.5	-8.5	0.4	
8	WIND LOAD +Z	-0.7	15.3	11.5	1.1	-16.8	11.0	1.6	-14.8	7.9	-1.3	14.7	9.0	2.9	67.5	27.8	-2.2	-69.3	0.1	4.3	-52.6	-0.3	-4.8	52.3	23.7	
9	SEISMIC LOAD +X	0.5	1.6	0.3	1.1	2.5	-0.1	1.1	-2.5	0.1	0.6	-1.5	-0.3	1.1	1.3	-0.1	0.9	1.7	0.0	0.9	-1.7	0.0	0.8	-1.4	0.1	
10	SEISMIC LOAD +Z	0.0	1.5	0.9	0.0	-1.5	0.9	0.1	-1.5	0.7	-0.1	1.5	0.8	0.3	5.1	1.8	-0.3	-5.1	0.0	0.5	-5.3	0.0	-0.5	5.3	1.9	

NOTE: PIPE LOAD +X & PIPE LOAD+Z ARE THE HORIZONTAL LOADS GENERATED DUE TO PIPE STRESS. THESE TWO LOADS SHALL BE CONSIDERED ACTING SIMULTANEOUSLY IN OPERATING CONDITION.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	22/02/2023	AS-BUILT	--	STY-SYC-SHH	--	BDA/PRI/JPD	
		C01	18/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S	--	STY/SYC/SHH	--	BDA/PRI/JPD	
				COMMENT ON 14/07/2020 AND REVISED AS MKD.					
		A01	30/05/2020	ISSUED FOR INFORMATION	--	STY/SYC/SHH	--	BDA/PRI/JPD	

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DRAWING NUMBER	12126-60-01-CS-131091	REVISION	0
STATUS	AFD		
DATE	11/05/2020	DATE	12/05/2020
DATE	30/05/2020		
DRAWN	SGH	CHECKED	ABM
APPROVED	DS		
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OUTOTEC DRAWING NUMBER	900707__GAU01__AG22__MTL01__39839	DOCUMENT ID	OU100687717
REVISION	2		

DRAWN	--	--
CHECKED	STY/ SYC/ SHH	28/05/2020
ENGINEER	--	--
APPROVED	BDA/ PRI/ JPD	30/05/2020
EMPLOYER		
SCALE	1:30	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
OXYGEN PLANT
FOUNDATION LOAD DATA-COLD BOX AREA(2 OF 2)

DRAWING NUMBER MD-216-3980-EG-CV-GAD-V551-1475 REV D01